

The Number of Solutions of $x^3 = 1$ in a 3-Group

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Let G be a finite group and let p be a prime divisor of $|G|$. Let $T = \{x \in G \mid x^p = 1\}$.

It is known ([2]) that if G is not a p -group then $|T| \leq \frac{p|G|}{p+1}$ and this inequality

is, for Mersenne primes, best possible. For $p=2$ it is easy to check that if G is a 2-group not of exponent 2, then $|T| \leq \frac{3}{4}|G|$ and this bound is attained e.g. in the dihedral group of order 8. In this note we show that $|T| \leq \frac{7}{9}|G|$ when G is a 3-group not of exponent 3 and again this bound is best possible. This answers a question posed by MacHale. The negative solution to the Hughes H_p -problem ([3]) suggests that no result of this type (i.e. $|T| \leq k|G|$ where $k < 1$ (k independent of G)) is likely for G a p -group not of exponent p for $p > 3$.

The notation is standard (c.f. [1]).

Our result is:

Theorem. *Let G be a finite 3-group not of exponent 3 and let $T = \{x \in G \mid x^3 = 1\}$.*

Then $|T| \leq \frac{7|G|}{9}$.

This inequality is best possible, for let

$$G = \left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & c & 1 & 0 \\ d & e & f & 1 \end{pmatrix} \mid a, b, c, d, e, f \in Z_3 \text{ and } a + c + f = 0 \right\}.$$

Then G has order 243, exponent 9 and $|T| = 189 = \frac{7}{9}|G|$.

Proof of Theorem. We assume the theorem is false and let G be a minimal counter-example. Let N be a normal subgroup of G of order 3. Then G/N has exponent 3 since, otherwise,

$$|\{gN \mid g^3 \in N\}| \leq \frac{7}{9}|G/N|$$

forcing

$$|T| \leq \frac{7}{9}|G|,$$

a contradiction. Next, if $N_1 \neq N$ is a normal subgroup of G of order 3, then G can be embedded in $(G/N) \times (G/N_1)$ which has exponent 3, giving a contradiction. So N is the unique minimal normal subgroup of G . In particular $Z(G)$ is cyclic. Since $|T| > \frac{1}{2}|G|$, $Z(G)$ has exponent 3.

We thus have

I. $|Z(G)| = 3$ and $G/Z(G)$ has exponent 3.

By results of Levi and van der Waerden ([1], III, (6.5), (6.6)) we have

II. (a) For $x, y \in G$,

(i) $[x, y, y] \in Z(G)$

(ii) $(xy)^3 = x^3 y^3 [x, y^{-1}]^3 [x, y^{-1}, x] [x, y^{-1}, y^{-1}]$

(b) G has (nilpotency) class at most 4.

Now II(a)(ii), (b) imply that if $w \in [G', G]$, then $(wg)^3 = w^3 g^3$ for all $g \in G$. Hence, since $|T| > \frac{1}{2}|G|$,

III. $[G', G]$ has exponent 3.

Let $g, h \in G$. Using II(b), we have

$$1 = [g, h^3] = [g, h]^3 [g, h, h]^3.$$

Hence III implies

IV. $[g, h]^3 = 1$ for all $g, h \in G$.

Let $x \in G' \cap T$. By II(b), $[x, w, x] = 1$ for all $w \in G$. Let $Y = T \cap x^{-1}T$. Then $Y = \{y \in T \mid [x, y, y] = 1\}$. Let $w \in T - Y$. Then xw and x^2w belong to $G - T$. Also if $w \in T - Y$, $xw \notin T - Y$, so $x(T - Y) \cap x^2(T - Y)$ is empty. Thus $|T - Y| \leq \frac{1}{2}|G - T| < \frac{1}{9}|G|$. Hence $|Y| > \frac{2}{3}|G|$. Now let $y \in Y$ and let $U = Y \cap y^{-1}Y$. Note that $|U| > \frac{1}{3}|G|$. Let $u \in U$. Then $[x, u, y] [x, y, u] = 1$.

The map: $g \rightarrow [x, g, y] [x, y, g]$ ($g \in G$) is a homomorphism of G whose kernel contains U . Hence $[x, g, y] [x, y, g] = 1$ for all $g \in G$.

For fixed $g \in G$, the map: $w \rightarrow [x, g, w] [x, w, g]$ ($w \in G$) is a homomorphism with Y in its kernel, so it is the identity. Hence we have

V. For $x \in G' \cap T$, $g, h \in G$, $[x, g, h] [x, h, g] = 1$.

In particular

VI. For $x \in G' \cap T$, $g \in G$, $[x, g, g] = 1$.

Now IV, VI and II(a)(i) imply that G satisfies the hypotheses of a theorem of Heineken ([1], III (6.9)), so

VII. G has class at most 3.

Now let $g \in G$ and let $h \in T \cap gT \cap g^2T$. Then, by II(a)(ii), $[g, h^{-1}, g] = 1$. The map: $w \rightarrow [g, w, g]$ ($w \in G$) is a homomorphism, so

VIII. $[g, w, g] = 1$ for all $g, w \in G$.

But now II(a)(ii) implies $(uv)^3 = u^3 v^3$ for all $u, v \in G$ and we get a contradiction immediately.

References

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